



PROFESSIONAL SUMMARY

"As a motivated Computer Science & IoT student, I aim to leverage my expertise in IoT, Embedded Systems, Full-Stack Development, and Automation to design innovative solutions. I seek opportunities to contribute to impactful projects, enhance my technical skills, and grow professionally while adding value to the organization."

KEY EXPERTISE

Arduino, IoT, Sensors, React.js, Node.js, MongoDB, Express.js, HTML, CSS, JS, REST API, CRUD, Frontend, Backend, Java, MySQL

EDUCATION

Sagar Group of Institutions - SISTec

2023 - 2027

B.Tech. - Internet of Things - SISTec-E | CGPA: 8.10 / 10

Gurudev Public School, Bina

2023

12th | MPBSE | Percentage: 72.40 / 100

Dream India School, Bina

2021

10th | MPBSE | Percentage: 80.60 / 100

PROJECTS

Libraray-Management-system

09 Feb, 2026 - 30 Mar, 2026

Mentor: Dharmendra Narayan Jha | Team Size: 1

Key Skills: MongoDB, Express.js, React.js, Node.js, JavaScript, HTML, CSS, REST API, CRUD, Authentication, Frontend, Backend

Project Link: <https://github.com/Anjali9109-Thakur/Libraray-Management-system>

****Project Title:**** Library Management System (MERN Stack)

****Description:****

The Library Management System is a web-based application developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js). The main objective of this project is to efficiently manage library operations such as book records, user management, and issue/return of books in a digital and organized manner.

The system provides features for adding, updating, and deleting book records, managing student or user details, and tracking issued and returned books. It ensures that all data is stored securely in a MongoDB database and can be accessed in real-time through a user-friendly interface. The frontend is developed using React.js, providing a responsive and interactive user experience, while the backend is handled using Node.js and Express.js to manage APIs and server-side logic.

The application implements CRUD operations for managing data and uses RESTful APIs for communication between frontend and backend. It also includes basic authentication mechanisms to ensure secure access to the system. The system reduces manual work, improves accuracy, and makes the process of managing library records faster and more efficient.

****My Role:****

This was an individual project, and I was responsible for designing and developing the complete system. I worked on frontend development using React.js and created responsive UI components. I also developed backend APIs using Node.js and Express.js, and managed the database using MongoDB. Additionally, I implemented CRUD operations, handled API integration, and performed testing and debugging to ensure smooth functionality of the application.

****Outcome:****

The project successfully digitizes library management processes and improves efficiency by reducing manual errors and saving time. It provides an easy-to-use interface for managing books and users, and ensures secure and organized data handling. The system demonstrates practical implementation of full-stack web development using the MERN stack and can be further enhanced with additional features in the future.

Smart Home Automation System

20 Aug, 2025 - 02 Mar, 2026

Mentor: Dharmendra Narayan Jha | Team Size: 5

Key Skills:

Arduino, Raspberry Pi, Python, C/C++, IoT, Sensors, Relay, Home Automation, Wi-Fi, MQTT, Embedded, Real-time, Automation

The Smart Home Automation System is a project designed to control and monitor home appliances remotely using IoT technology. The system allows users to operate lights, fans, and other devices through a mobile application or web interface, providing convenience, energy efficiency, and enhanced security.

The project uses Arduino and Raspberry Pi as core controllers. Various sensors such as temperature, motion, and light sensors are integrated to automate device operations based on environmental conditions. Devices are connected through Wi-Fi and controlled using protocols like MQTT for real-time monitoring and automation. Python and C/C++ are used for programming the controllers and implementing automation logic.

My Role:

I was responsible for designing and implementing the system architecture, integrating sensors, programming the controllers, and ensuring smooth communication between devices and the control interface. I also tested and optimized the system to ensure real-time responsiveness and reliability.

Outcome:

The system successfully automates home appliances, reduces manual effort, and improves energy efficiency. It demonstrates practical implementation of IoT and embedded systems, providing a scalable and secure solution for smart home management.

Attendance-system-by-image-detection-and-face-recognition

10 Sep, 2025 - 02 Feb, 2026

Mentor: Dharmendra Narayan Jha | **Team Size:** 5

Key Skills: Python OpenCV Flask Haar Cascade Classifier MongoDB HTML CSS JavaScript NumPy

Project Link: <https://github.com/Anjali9109-Thakur/Attendance-system-by-image-detection-and-face-recognition>

****Project Title:**** Attendance System using Image Detection and Face Recognition

****Description:****

This project focuses on developing an automated attendance management system using image detection and face recognition technology. The main objective of the system is to eliminate the need for manual attendance marking and provide a fast, accurate, and contactless solution. The system captures real-time images through a camera, detects faces using Haar Cascade Classifier, and recognizes individuals by comparing facial features with a pre-trained dataset. Once a face is recognized, the system automatically marks attendance and stores the data in a database.

The project is implemented using Python as the core programming language. OpenCV is used for image processing and face detection, while NumPy is utilized for handling numerical data. Flask is used to develop a simple web-based interface, and MongoDB is used for storing user details and attendance records. The frontend is designed using HTML, CSS, and JavaScript to provide a user-friendly interface.

****My Role:****

I was responsible for handling the database part of the project. My role included designing and managing the database using MongoDB, storing user details and attendance records, and ensuring efficient data retrieval and storage. I worked on integrating the database with the backend using Flask and Python, and ensured that attendance data was accurately recorded and maintained. I also handled data organization, updating records, and maintaining data consistency throughout the system.

****Outcome:****

The system successfully automates the attendance process with improved accuracy and efficiency. It reduces human effort, minimizes errors, and saves time compared to traditional methods. The project demonstrates practical implementation of computer vision and machine learning concepts in a real-world application. It also enhances security by ensuring that attendance is marked only for authorized individuals. Overall, the project provides a reliable and scalable solution for attendance management in educational institutions and organizations.

ACHIEVEMENTS

- o Successfully developed and deployed Smart Home Automation System
- o Developed Library Management System (MERN Stack) as an individual project for efficient book management.
- o Contributed to Attendance System using Face Recognition as a team project, implementing database integration and backend logic.

ASSESSMENTS / CERTIFICATIONS

Introduction to Internet of Things (IoT)

Provider: NPTEL

Aggregate: 66 / 100

Subjects: IoT (66 / 100)

Key Skills: IoT, Sensors, Embedded Systems, Networking, Data Analysis

Completed NPTEL course on Introduction to IoT, learning concepts of sensors, embedded systems, networking, and data analysis for IoT applications.

Object-Oriented Programming (OOPS)

Provider: NPTEL

Aggregate: 60 / 100

Subjects: OOPS (60 / 100)

Key Skills: OOPS, Java, Classes, Objects, Inheritance, Polymorphism, Programming

Completed NPTEL course on Object-Oriented Programming, mastering concepts of classes, objects, inheritance, polymorphism, and software development best practices.

National Aeromodelling Workshop

Provider: IIT Kanpur / Boeing

Key Skills: Aeromodelling, Aircraft Design, RC Aircraft, Flight Mechanics, Engineering Design, Teamwork, Problem Solving

Attended 2-day National Aeromodelling Workshop hosted by IIT Kanpur and Boeing, learning design, fabrication, and testing of RC aircraft, gaining hands-on skills in aerodynamics, flight mechanics, and teamwork.

SISTEC Innovation Hackathon 3.0

Provider: SISTEC

Key Skills: Innovation, Problem Solving, Teamwork, Project Development

Participated in SISTEC Innovation Hackathon 3.0, working on innovative projects with a team, developing problem-solving, coding, and teamwork skills in a competitive environment.

Web Development: HTML, CSS, JavaScript

Provider: Cisco Networking Academy

Key Skills: HTML, CSS, JavaScript, Web Development, Frontend Development, Programming

Completed Cisco Networking Academy Web Development course, learning core HTML, CSS, and JavaScript concepts for creating responsive web pages and basic frontend applications.

POSITIONS OF RESPONSIBILITY

I am a Member of Student Activity Council (SAC) in College, 2025–2026

Represented students and coordinated college activities and events.

Assisted in planning and managing cultural, technical, and social programs.

Developed strong **leadership, teamwork, and organizational skills**.

Corporate Crew in Training & Placement Department

Assisted in organizing placement drives, workshops, and training sessions for students.

Coordinated between companies and students to ensure smooth scheduling and communication.

Developed strong **communication, coordination, and problem-solving skills**.

SEMINARS / TRAININGS / WORKSHOPS

National Aeromodelling Workshop Institute Name: IIT Kanpur

11 Jan, 2025 - 12 Jan, 2025

Key Skills: Aeromodelling, Aircraft Design, RC Aircraft, Flight Mechanics, Engineering Design, Teamwork, Problem Solving

Attended a 2-day National Aeromodelling Workshop at IIT Kanpur focused on the design, fabrication, and testing of radio-controlled (RC) aircraft. Learned concepts of flight mechanics, aerodynamics, and aircraft control systems, along with hands-on experience in building and testing model planes. Gained skills in teamwork, problem-solving, and technical design, applying theoretical knowledge to practical projects.

CO-CURRICULAR ACTIVITIES

- Member, Student Activity Council (SAC), 2025–2026 & Corporate Crew, Training & Placement Department,

EXTRA CURRICULAR ACTIVITIES

- Participated in Inter-College Hackathons, showcasing skills in coding, problem-solving, and project development. & Organized college cultural and tech events

PERSONAL INTERESTS / HOBBIES

- Reading technical and script books Learning new technologies

WEB LINKS / IMs

- o Other - https://www.linkedin.com/in/anjali-lodhi-03972832b?utm_source=share_via&utm_content=profile&utm_medium=member_android
- o Github - <https://github.com/Anjali9109-Thakur>

PERSONAL DETAILS

Gender: Female

Marital Status: Single

Current Address: Neelbad Bhopal, Sehore Road Bhopal, Bhopal, Madhya Pradesh, India - 462044

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Date of Birth: 19 Nov, 2005

Known Languages: Hindi & English

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